

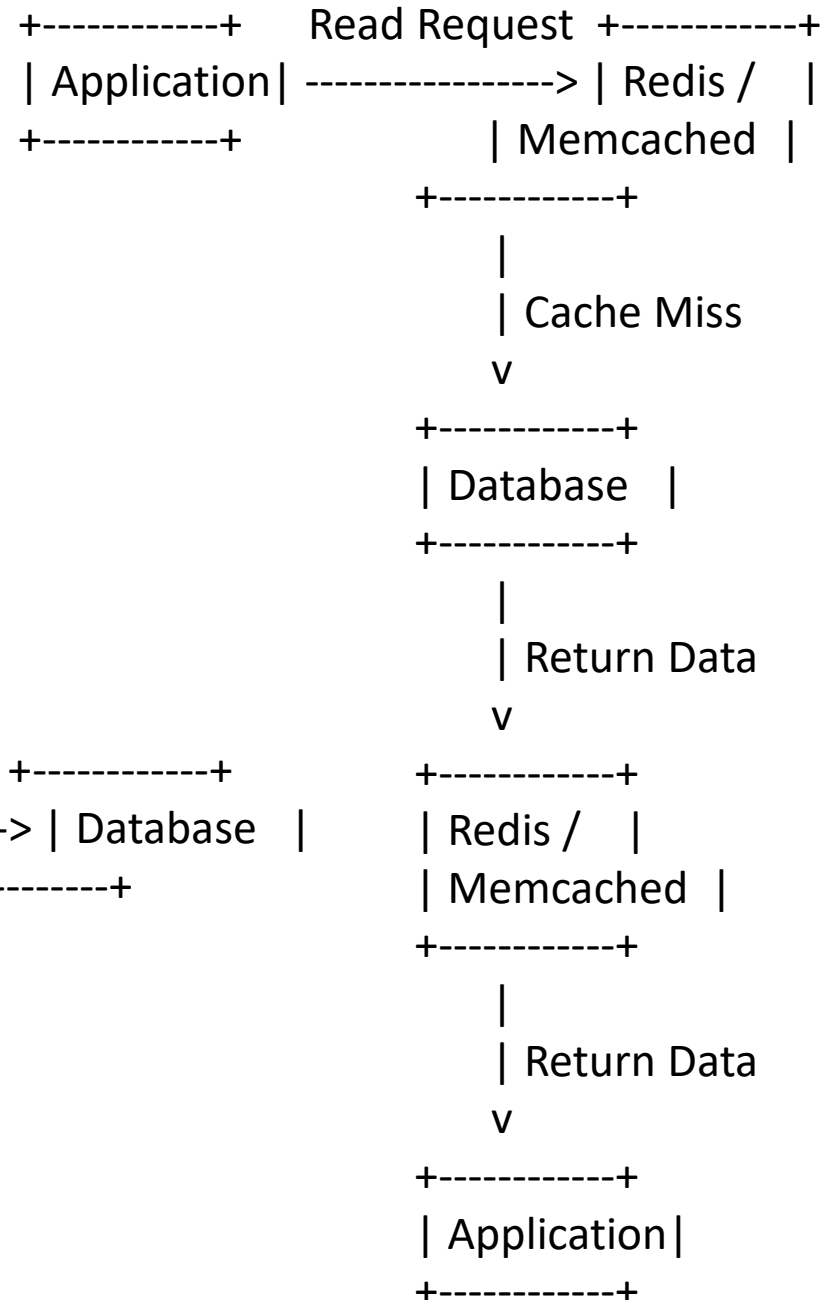
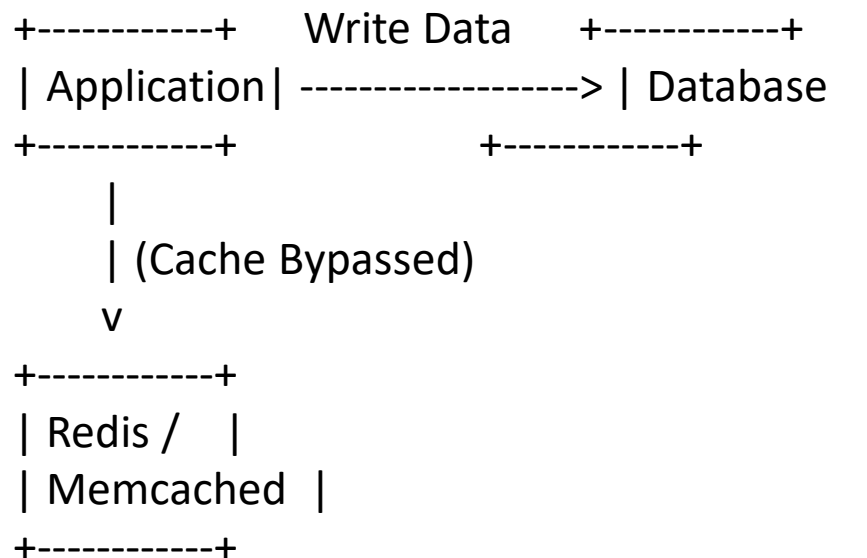
# Caching Strategies Part 5

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# Write-Around Overview

- **Write-Around** means data is written **directly to the database**, bypassing the cache.
- The cache is **updated only when the data is read** later.

## WRITE OPERATION

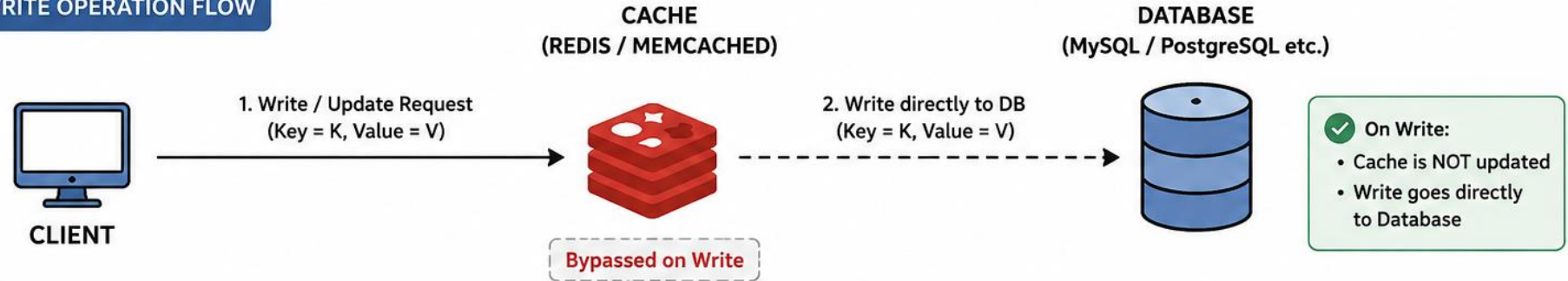


# Working

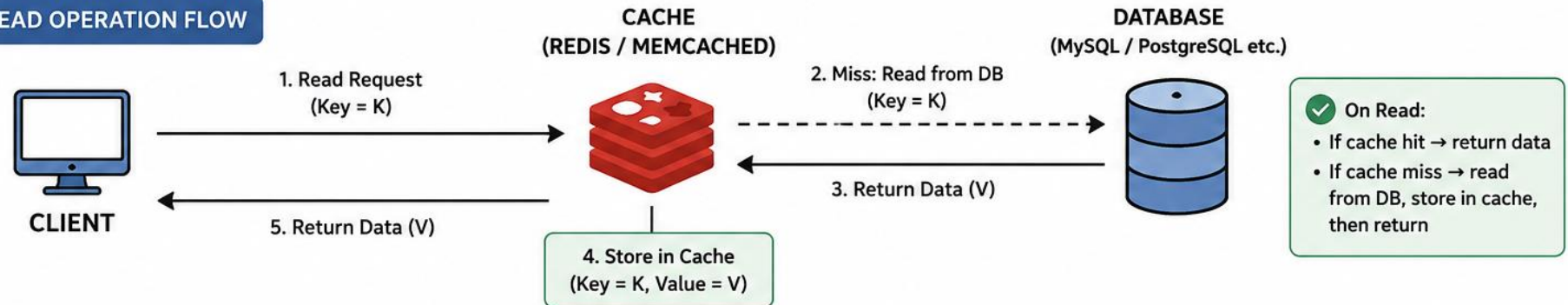
## Write Around Caching in Redis / Memcached

**Write Around means:** On write, data is written directly to the database (cache is bypassed).  
Cache is only updated when the data is read.

### 1. WRITE OPERATION FLOW



### 2. READ OPERATION FLOW



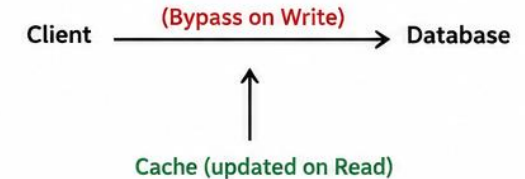
#### KEY POINTS

- Writes go around the cache (directly to DB).
- Cache is updated only when data is read (lazy loading).
- Reduces cache pollution from write-heavy workloads.
- Ensures cache contains only frequently read data.



#### Use Case

- Ideal for systems with:
- High write frequency
  - Read data is a small subset
  - Expensive cache updates



# Advantages and Disadvantages

- **Advantages**

- ✓ Reduces cache pollution (rarely-read data isn't cached)
- ✓ Better cache utilization
- ✓ Good for write-heavy workloads
- ✓ Lower cache write traffic

- **Disadvantages**

- ✗ First read after a write is slower (cache miss)
- ✗ More database reads
- ✗ Higher read latency for newly written data

# Best Use Cases

- Logging systems
- Analytics platforms
- IoT data ingestion
- Applications with very high write volume and relatively low read frequency

# Comparison

| Strategy      | Write Path               | Cache Updated on Write? |
|---------------|--------------------------|-------------------------|
| Cache-Aside   | Database                 | No                      |
| Write-Through | Cache → Database         | Yes                     |
| Write-Back    | Cache → Database (async) | Yes                     |
| Write-Around  | Database only            | No                      |

In Redis or Memcached, **Write-Around** is usually implemented in the **application layer** because neither cache natively provides automatic write-around behavior.

- Thank You